Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/21/2014
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Utilizing the Advanced Biological Research System (ABRS) on the International Space Station (ISS) to Uncover Microgravity's Impact on Root Development and Cell Wall Architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
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PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/25/2012	**End Date:**	07/24/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX12AM94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this proposal is to uncover the underlying molecular mechanisms by which microgravity impacts cell wall architecture in the model plant Arabidopsis thaliana using the Advanced Biological Research System (ABRS) housed on the International Space Station (ISS). This proposal builds on a robust dataset from our rapid turnaround spaceflight experiments, which utilized the Biological Research in Canisters (BRIC16) hardware on the STS-131 mission. Through experiments with the BRIC hardware, we discovered that spaceflight had a profound influence on the expression of genes involved in regulating cell wall architecture. This led us to hypothesize that plants respond to microgravity in large part through the transcriptional reprogramming of genes that control cell wall remodeling. Because of the prominent root developmental phenotypes (i.e. primary root skewing and inhibition of root hair growth) that we observed in our BRIC experiments, it appears that microgravity-dependent changes in cell wall-related transcripts are predominantly manifested in roots. Furthermore, because primary root and root cell wall ultrastructural defects were most pronounced in a mutant to a vegetative actin isoform (act2), we hypothesize that microgravity predominantly affects genes that lie within signaling networks linking the actin cytoskeleton with the plant extracellular matrix. Given our successful experiments with BRIC, we are in an excellent position to effectively utilize the unique capabilities of the ABRS hardware on the ISS to test the aforementioned hypotheses. We will use a new algorithm that we developed from our BRIC16 experiments to more rigorously quantify root growth behavior of wild-type and a higher order vegetative actin mutant (act2 act8 double mutant) (aim 1). This will clarify the extent of actin involvement in specifying root orientation and cell wall remodeling in space. Through the use of monoclonal antibodies and immunocytochemistry, we will also determine how the distribution and abundance of specific cell wall components change in microgravity, and correlate such changes with root developmental responses and modifications in gene expression (aim 2). Finally, we will employ more sensitive deep sequencing technologies (e.g. RNA-Seq) to discover novel microgravity-induced transcriptional changes in plants (aim 3). Knowledge from our spaceflight studies will then be translated into corresponding gene functional studies (e.g. mutant analysis and overexpression), which we expect will lead to the discovery of new players that modulate root development on earth and in microgravity. The proposed work is therefore relevant to this NASA-NRA because it will address fundamental questions on how gravity modulates the plant's internal cytoarchitecture and the molecular processes that govern the development of its cell walls. This will be significant for NASA's mission as extrapolated in the priorities of the Fundamental Space Biology Plan for 2010-2020 because it will provide us with basic knowledge to guide genetic engineering efforts to generate plants that are better adapted to the weightless environment of spaceflight. This in turn will be beneficial for the design of large plant habitats on the ISS and future space colonies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a deeper understanding of how microgravity influences plant development. The focus of this research will be on the development of roots and how they modify the expression of their genes to adapt to the microgravity environment of spaceflight. This research will address one of high priority recommendations of the National Research Council's 2011 decadal survey report, which is to contribute significantly to building a robust spaceflight program to analyze plant physiological responses to stimuli encountered in microgravity. Research areas depicted in this proposal, namely root development and cell wall biology, are significant for NASA's goal of developing large plant habitats on the International Space Station (ISS) and future space colonies. We expect that information gained from basic studies of root growth in microgravity will guide the development of plant varieties better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Furthermore, because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of its cell walls, this research will provide a deeper mechanistic understanding of the genetic networks that specify how cell walls are constructed in microgravity. In space, such knowledge will enable technologies for designing plants that can sustain optimal growth and stature despite the adverse impact that microgravity has on cell wall development. With regard to Earth benefits, understanding of molecular pathways that govern root growth and cell wall development, will benefit agricultural practices and bioenergy research on Earth.
Task Progress:	For the first and second year of this grant, we completed both the Science Verification Test (SVT) and the Payload Verification Test (PVT). The SVT was held at the Kennedy Space Center (KSC) in October, 2012 while the PVT was conducted in March, 2013, also at KSC. Results from the APEX 02-1 SVT and PVT were submitted for peer review this year and the manuscript is currently in press with the journal, Gravitational and Space Research (Nakashima et al., 2014b). Some of the results that will be published in Gravitational and Space Research are briefly presented in below. SVT and PVT focused on two success criteria. First, we evaluated a system to delay Arabidopsis seed germination until installation on ABRS. Second, we assessed the quality of seedling fixation on Kennedy Fixation Tubes (KFTs) for post-flight microscopic analyses. The former criterion was important to ensure that seedling development occurred only in microgravity and therefore minimized any additional source of variation that could complicate the interpretation of science results. Furthermore, delaying seed germination until the experiment was activated on ABRS would eliminate the need for replanting in case the launch was postponed by several days or weeks. The ability to delay seed germination could also provide the ISS crew flexibility in regard to timing Petri plate installation on ABRS. SVT and PVT led to the design of a portable metal box for FR illumination that would enable us to delay Arabidopsis seed germination until installation on ABRS. The short FR light treatment did not compromise seed viability and could potentially delay germination for up to 10 weeks in the Arabidopsis Col-0 ecotype. Through SVT and PVT, we have also determined an optimal fixative composition as well as additional ways to handle fixative prior to loading on KFTs. Flexibility in conducting space biology experiments is often dictated by hardware design and established operating procedures specified by NASA because of safety and other logistical issues. Lessons learned from SVT and PVT, which typically are not shared with the scientific community, can be valuable to increase the likelihood that the quality of plant material obtained from spaceflight is not compromised. Unfortunately, after completing SVT and PVT with the ABRS, the ABRS hardware on the ISS experienced technical issues in regard to the control of specific environmental parameters. It was determined that these issues could not be resolved in time for SpaceX-3 launch. Thus, we were de-manifested from SpaceX-3 and we were informed of the need to use an alternative hardware. The hardware chosen for implementation of our spaceflight experiments is the vegetable production unit (VEGGIE). The new experiments using VEGGIE are now called APEX03-1 (original experiment was APEX02-1). SVT for APEX03 is scheduled for June, 2014. Prior to SVT we completed preliminary testing of the VEGGIE hardware in February, 2014. We found that VEGGIE is able to support robust growth of Arabidopsis

	seedlings, which should still allow us to accomplish the original goals of APEX02-1. However, one drawback of using VEGGIE is that we can no longer obtain root growth measurements with increased temporal resolution. Root images will only be obtained at two time points (i.e. during the 7 day and 14 day harvest).
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	Nakashima J, Sparks JA, Carver JA Jr, Stephens SD, Blancaflor EB. "Preparations for Advanced Plant EXperiments (APEX) 02-1 in space using the Advanced Biological Research System (ABRS)." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 120. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Kwon T, Sparks JA, Nakashima J, Allen S, Tang Y, Blancaflor EB. "Long-term exposure to microgravity triggers transcriptional reprogramming of cell wall-related genes in Arabidopsis." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 70. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Sparks JA, Blancaflor EB. "A Mitochondria-Localized Tetratricopeptide Repeat-Like Superfamily Protein Functions in Actin-Mediated Root Developmental Processes." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 122. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Articles in Peer-reviewed Journals	Nakashima J, Liao F, Sparks JA, Tang Y, Blancaflor EB. "The actin cytoskeleton is a suppressor of the endogenous skewing behaviour of Arabidopsis primary roots in microgravity." Plant Biology (Stuttg). 2014 Jan;16 Suppl 1:142-50. Epub 2013 Aug 16. http://dx.doi.org/10.1111/plb.12062 ; PubMed PMID: 23952736 , Jan-2014
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Articles in Peer-reviewed Journals	Dyachok J, Sparks JA, Liao F, Wang YS, Blancaflor EB. "Fluorescent protein-based reporters of the actin cytoskeleton in living plant cells: fluorophore variant, actin binding domain, and promoter considerations." Cytoskeleton (Hoboken). 2014 May;71(5):311-27. http://dx.doi.org/10.1002/cm.21174 ; PubMed PMID: 24659536 , May-2014